AI-Assisted EDA Pipeline

GDOT Traffic Dataset -- Subagent Workflow Design

How the EDA was decomposed into eight specialized analysis agents

8	7	286	818K
Specialized Subagents	Analysis Tracks	Stations Analyzed	Clean Records Produced

This document describes how the GDOT Traffic Dataset EDA was structured as a multi-agent AI pipeline. Rather than running a single monolithic analysis, the project was decomposed into eight specialized subagents -- each with a defined scope, a set of analytical questions, and concrete deliverables.

The pipeline was designed so each agent builds on the previous: starting from raw data inventory, progressing through quality auditing and behavioral profiling, and culminating in a modeling-ready feature dataset and synthesized EDA report.

Contents: (1) Design rationale (2) Pipeline architecture (3) Agent descriptions (4) Outputs and oversight

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1. Why a Multi-Agent Pipeline?

The GDOT dataset spans three data tiers, 286 stations, and eight years of traffic records. Decomposing it into eight focused agents ensured each domain was covered thoroughly and independently.

- **Separation of concerns:** Each agent owns one question type -- quality, behavior, trends, etc. -- without mixing concerns.
- **Parallelism:** Independent agents (traffic profiling, corridor analysis) can run simultaneously.
- **Auditability:** Every agent saves outputs separately, making any finding traceable back to its source.
- **Scalability:** Scripts run across all 286 stations; adding new years only requires re-running the relevant agent.
- **Manager synthesis:** A final manager agent reads all outputs, resolves contradictions, and produces the unified EDA report.

2. Pipeline Architecture

Agents 1-2 must finish first (trust flags from the quality audit feed downstream). Agents 3-6 can run in parallel. Agent 7 requires 3-6 to complete. Agent 8 synthesizes all outputs.

Note: Agents 1-2 are sequential prerequisites. Agents 3-6 are parallel. Agent 7 requires agents 3-6. Agent 8 reads all outputs and is the final step.

3. Agent Descriptions

1

Understand what data exists before any analysis begins

Total station count, station-years, station-months	Station master table (286 stations)
Functional class distribution (Interstate to Collector)	Completeness table by station-month
Urban vs. rural breakdown	Metadata summary and coverage heatmap
Complete vs. incomplete month counts per station	Schema / data dictionary audit
Schema consistency across data tiers and years	

RESULT: format mismatch flagged for downstream handling).

2

Quantify and characterize all data gaps before behavioral analysis

% missing by station, month, hour, day of week	Missingness scorecards per station
Station x date missingness heatmaps	Outage pattern taxonomy
Run-length distribution of consecutive missing days	Flagged data-quality windows
Zero-volume and low-coverage interval detection	Cautious MCAR/MAR language enforced throughout
Outage pattern labeling: random / block / bursty / weekday-h	

RESULT: methodology differences rather than sensor failure.

3

Establish what normal traffic looks like at each station

	KEY DELIVERABLES
Daily volume distributions per station	Station profile plots (hourly, DOW, monthly)
Hourly volume profiles and AM/PM peak times	AM/PM peak time summary across all stations
Day-of-week and monthly seasonal profiles	Healthy traffic signature table
Weekday vs. weekend ratio comparisons	Station archetype notes by functional class
Vehicle mix, speed, and class distributions	

RESULT: confirmed -- reliable fixed features for AADT models.

4

Understand how traffic evolved over 8 years including COVID effects

	KEY DELIVERABLES
Annual AADT trends by station (2017-2024)	AADT trend plots and summary table
Year-over-year monthly volume changes	Decomposition plots per station
STL seasonal decomposition (trend + seasonality + residual)	Holiday effect ratios (6 holidays x 5 functional classes)
Holiday and long-weekend effect quantification	Long-weekend day-by-day volume ratio chart
AM/PM peak hour factor stability across years	

RESULT: correction factors computed for all functional classes.

5

Identify which stations behave similarly for gap-filling imputation

	KEY DELIVERABLES
Pairwise Pearson correlation matrices by month	Pairwise correlation matrix (286 x 286)
Station-pair daily volume scatter plots	Behavioral neighbor table per station
Lead/lag cross-correlation between stations	27 reference station candidates (r > 0.85)
Distance vs. correlation relationship	Corridor relationship memo

Behavioral vs. geographic neighbor comparison

RESULT: corridor boundaries in several cases.

6

Flag suspicious periods and assign trust tiers to every station-month

	KEY DELIVERABLES
Anomaly detection: rolling median + MAD method	Anomaly flags with event attribution
Anomaly detection: daily IQR-based thresholds	Station trust tiers: TRUSTED / CAUTION / EXCLUDED
Sensor-error anomalies vs. real-world events	Station-month representativeness assessment
Representativeness of incomplete station-months	Event impact summary (90.5% of anomalies explained)
Observed-day sufficiency analysis	

RESULT: all quality checks.

7

Produce the final modeling-ready dataset with engineered features

	KEY DELIVERABLES
Compare 5-min / 15-min / hourly / daily resolutions	clean_hourly_2024_trusted.csv -- 818,645 records
Engineer temporal features (DOW, month, peak flags, season)	286-station feature table with 27 predictors
Create event flags (holiday, construction, long weekend)	all_events_2024.csv -- event flag database
Build lag and rolling volume features	Feature specification and preprocessing document
Composition features (truck share, heavy vehicle share)	

RESULT: ready for regression pipeline.

8

Synthesize all agent outputs into one coherent EDA conclusion

	KEY DELIVERABLES
Cross-agent contradiction resolution	Master EDA report (PDF, 14 pages)
Strongest finding identification with evidence	Presentation deck (PPTX, 15 slides)
Risk and caveat documentation	Trusted stations list (243) + caution list (36)
Modeling readiness assessment	Modeling readiness statement and next-step plan
Trusted and caution station list compilation	

RESULT: missingness ambiguity, 5 sparse stations requiring field record investigation.

4. Key Pipeline Outputs

Together, the eight agents produced the following concrete deliverables that form the complete EDA record for the GDOT project:

clean_hourly_2024_trusted.csv	818,645 hourly records, 243 trusted stations
station_catalog.csv	286-station master table with metadata and trust tier
all_events_2024.csv	Holiday, construction, and weather event flag database
Feature table (27 predictors)	DOW, month, peak flags, rolling volumes, truck share
Missingness scorecards	Per-station % missing, outage pattern, quality flags
Holiday effect ratios	6 holidays x 5 functional classes -- correction factors
27 reference stations	Behavioral neighbors for gap-filling imputation
Trust tier assignments	243 TRUSTED / 36 CAUTION / 7 EXCLUDED
Master EDA report (PDF, 14 pages)	Full narrative findings with all figures
Presentation deck (PPTX, 15 slides)	Story-arc slides with speaker notes and callouts

5. Human Oversight and Verification

The subagent pipeline accelerated the analysis but did not replace human judgment. Every stage involved direct review:

- **Code review:** All agent-produced scripts were read, tested, and run manually.
- **Output verification:** Every figure was reviewed -- several required targeted debugging (e.g., holiday figures blank due to a COSIT ID format mismatch).
- **Trust decisions:** Station tier assignments were cross-checked against known events and domain knowledge before being accepted.
- **Language caution:** Missingness mechanism language was revised to remain cautious -- "consistent with structured missingness" not "MAR" or "MCAR".
- **Presentation edits:** Slide titles, narrative framing, and speaker notes were all edited by the analyst after reviewing each agent draft.

AI agents handled decomposition, code drafting, and output organization. The analyst handled verification, interpretation, and all final decisions. The pipeline made it possible to cover 286 stations across 8 years thoroughly -- work that would have taken significantly longer without structured agent support.

8 agents | 286 stations | 818K records | Data ready for AADT modeling